

Lab - Penalty method

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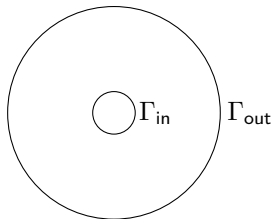
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May 23, 2025

PhD Course - *Numerical Methods for Multi-Physics Problems in Moving Domains*

Find the solution $u : \Omega \times (0, T] \rightarrow \mathbb{R}$ (e.g. temperature) such that

$$\left\{ \begin{array}{ll} \partial_t u - \operatorname{div}(\mu \nabla u) = f & \text{in } \Omega, t \in (0, 2] \\ u = 0 & \text{on } \Gamma_{\text{ext}}, t \in (0, 2] \\ u = 0.1 & \text{on } \Gamma_{\text{in}}, t \in (0, 2] \\ u = 0 & \text{in } \Omega, t = 0 \end{array} \right.$$



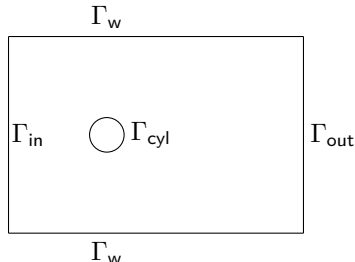
Starting from the template [TEMPLATE_HeatBrinkman.py](#), solve the problem with

- conforming boundary conditions
- Brinkman penalty method

Stokes equations

Find the solution $\mathbf{u} : \Omega \times (0, 10] \rightarrow \mathbb{R}^d$ and $p : \Omega \times (0, 10] \rightarrow \mathbb{R}$ such that

$$\left\{ \begin{array}{ll} \rho \partial_t \mathbf{u} - \operatorname{div} \boldsymbol{\sigma} = \mathbf{f} & \text{in } \Omega, t \in (0, 10] \\ \operatorname{div} \mathbf{u} = 0 & \text{in } \Omega, t \in (0, 10] \\ \mathbf{u} = \mathbf{u}_{\text{in}} & \text{on } \Gamma_{\text{in}}, t \in (0, 10] \\ \boldsymbol{\sigma} \mathbf{n} = \mathbf{0} & \text{on } \Gamma_{\text{out}}, t \in (0, 10] \\ \mathbf{u} = \mathbf{0} & \text{on } \Gamma_{\text{w}}, t \in (0, 10] \\ \mathbf{u} = \mathbf{u}_{\text{cyl}} & \text{on } \Gamma_{\text{cyl}}, t \in (0, 10] \\ \mathbf{u} = \mathbf{0} & \text{in } \Omega, t = 0 \end{array} \right.$$



with $\mathbf{u}_{\text{in}} = [4(1 - e^{-t})y(1 - y), 0]^T$

Starting from the template [TEMPLATE_StokesBrinkman.py](#), solve the problem with

- conforming boundary conditions
- Brinkman penalty method (with $\mathbf{u}_{\text{cyl}} = 0$)
- Brinkman penalty method with a cylinder moving with the prescribed kinematics $\mathbf{d} = [d, 0]^T = [A(1 - \cos(0.2\pi t)), 0]^T$